Power spectra analysis post-processing directions:

Using MATLAB and R

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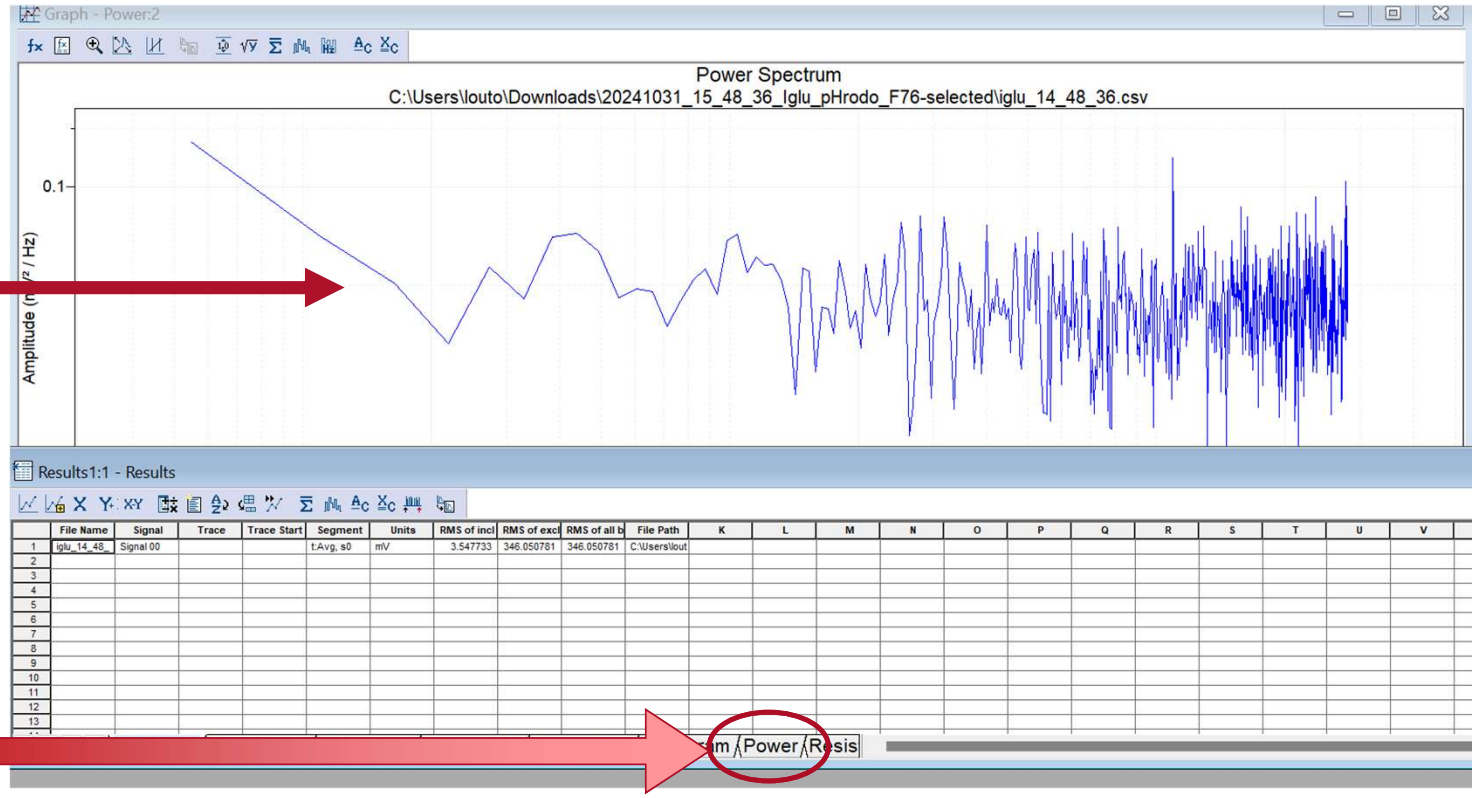
Exporting Clampfit data in the proper format

1. Load your ImageJ csv's for each cell in one scan into Clampfit

2. Follow the typical Analyze-Power Spectra-Append path to generate your spectra

3. In the results window, go to the "Power" tab

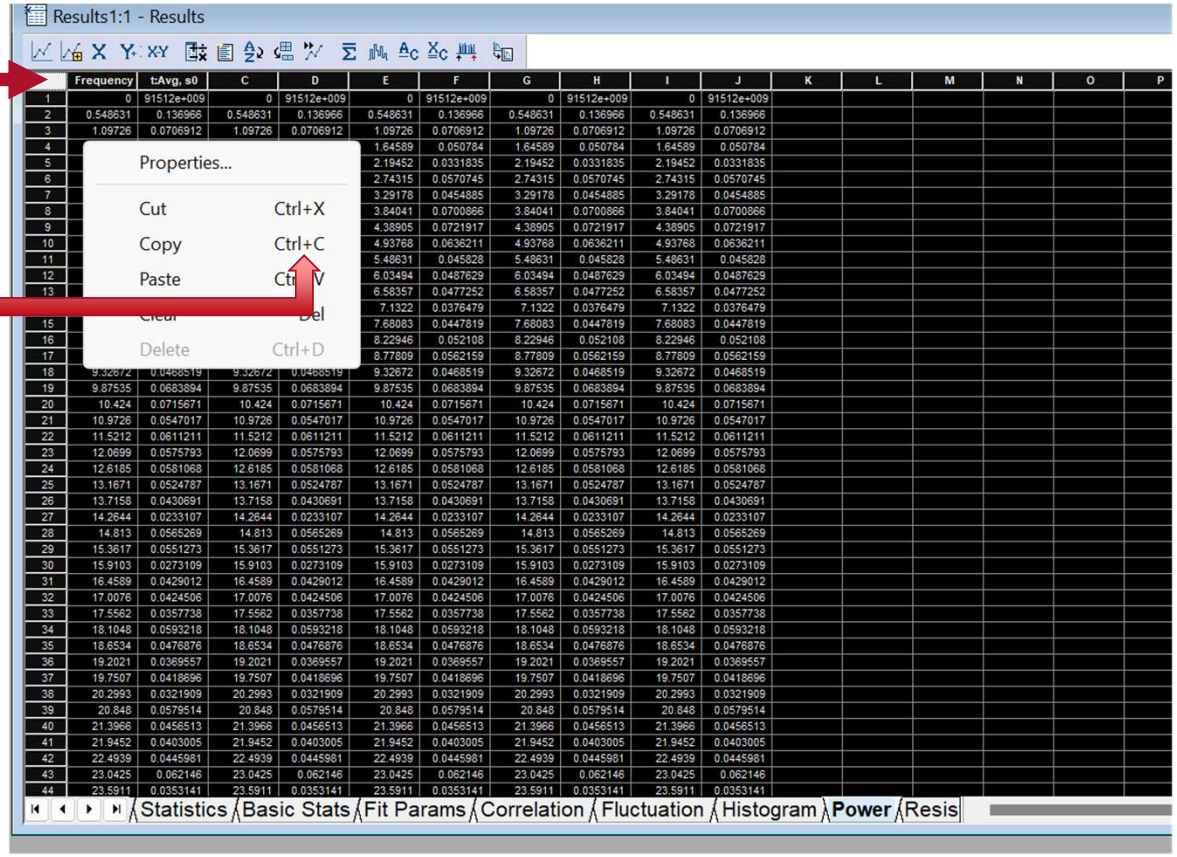


Exporting Clampfit data in the proper format

1. In this example of power spectral data from 5 cell traces, I have highlighted all the data by clicking the lefthand corner

2. Then I right-click or CTL+C to copy this data to be pasted into an .XLSX file in Excel

3. You do not need the “Frequency” and “tAvg;..” labels, only the data



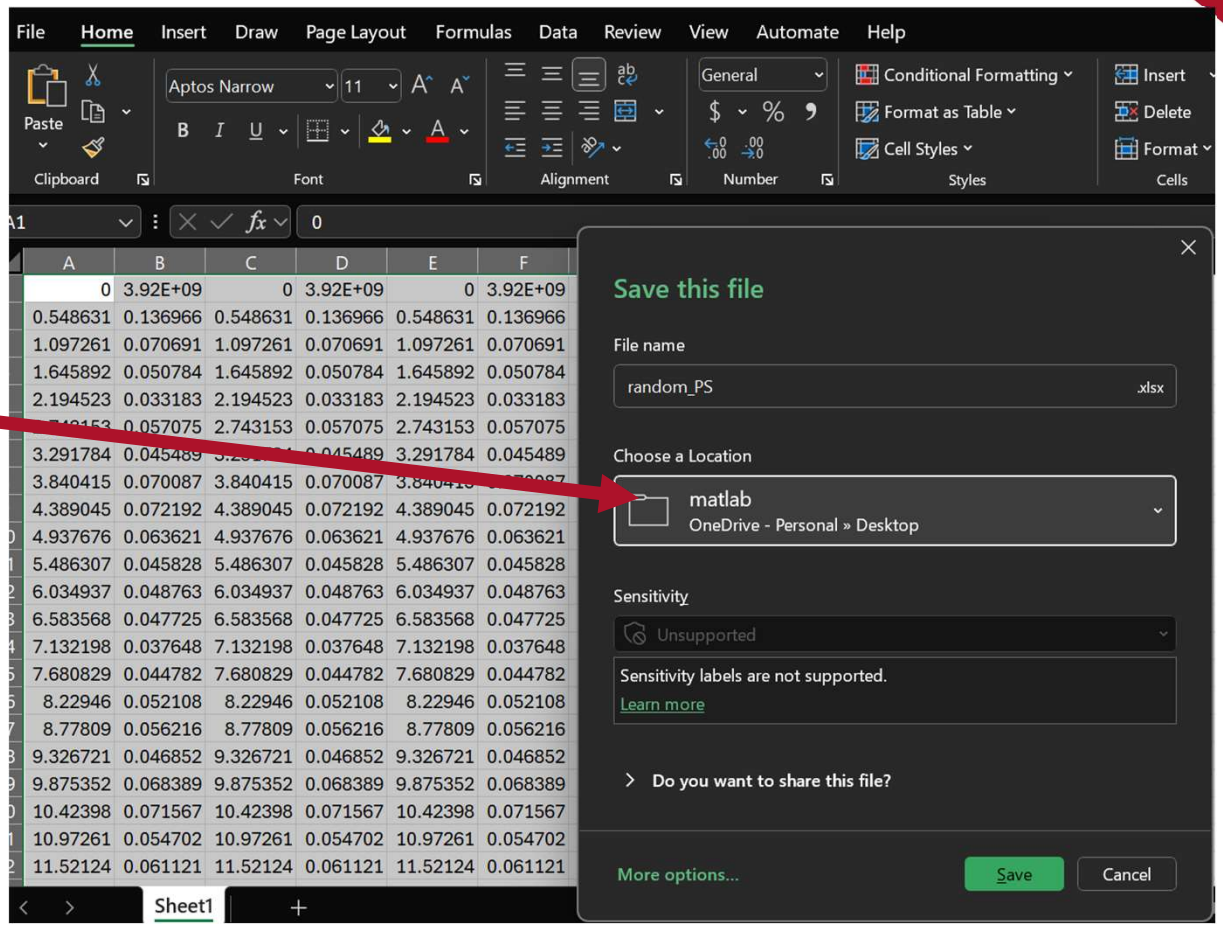
Exporting Clampfit data in the proper format

1. Paste the data in Excel and save it to the path you intend to work in to run the MATLAB script

   I made mine "random_PS.xlsx" for all examples in this module

2. For me, I like to work out of my "matlab" folder at all times, so I save all my data sheets of interest here

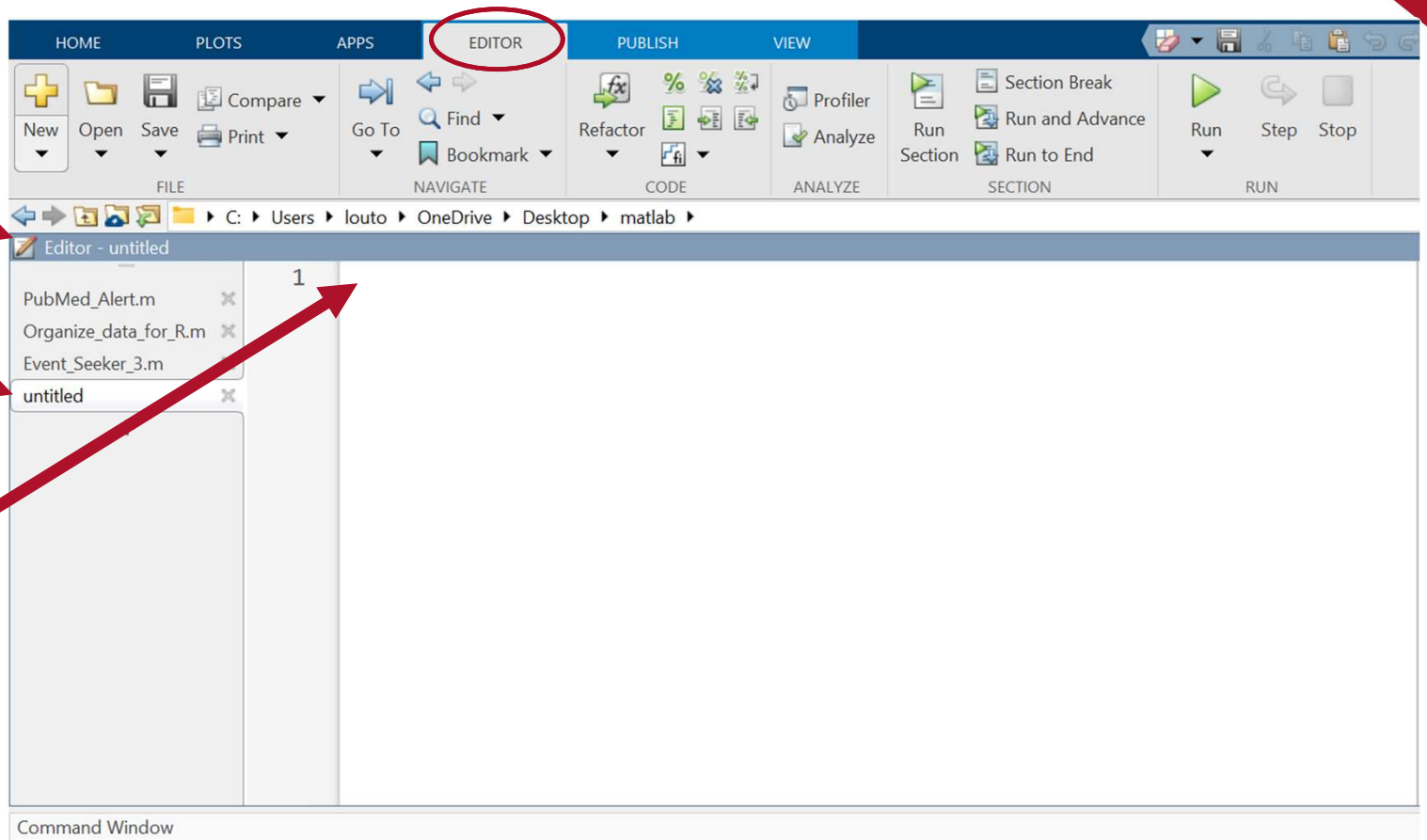
3. This folder is located in my OneDrive-> Desktop path (it is important to keep track of paths for later)



Using MATLAB and loading excel files containing data

The MATLAB Layout

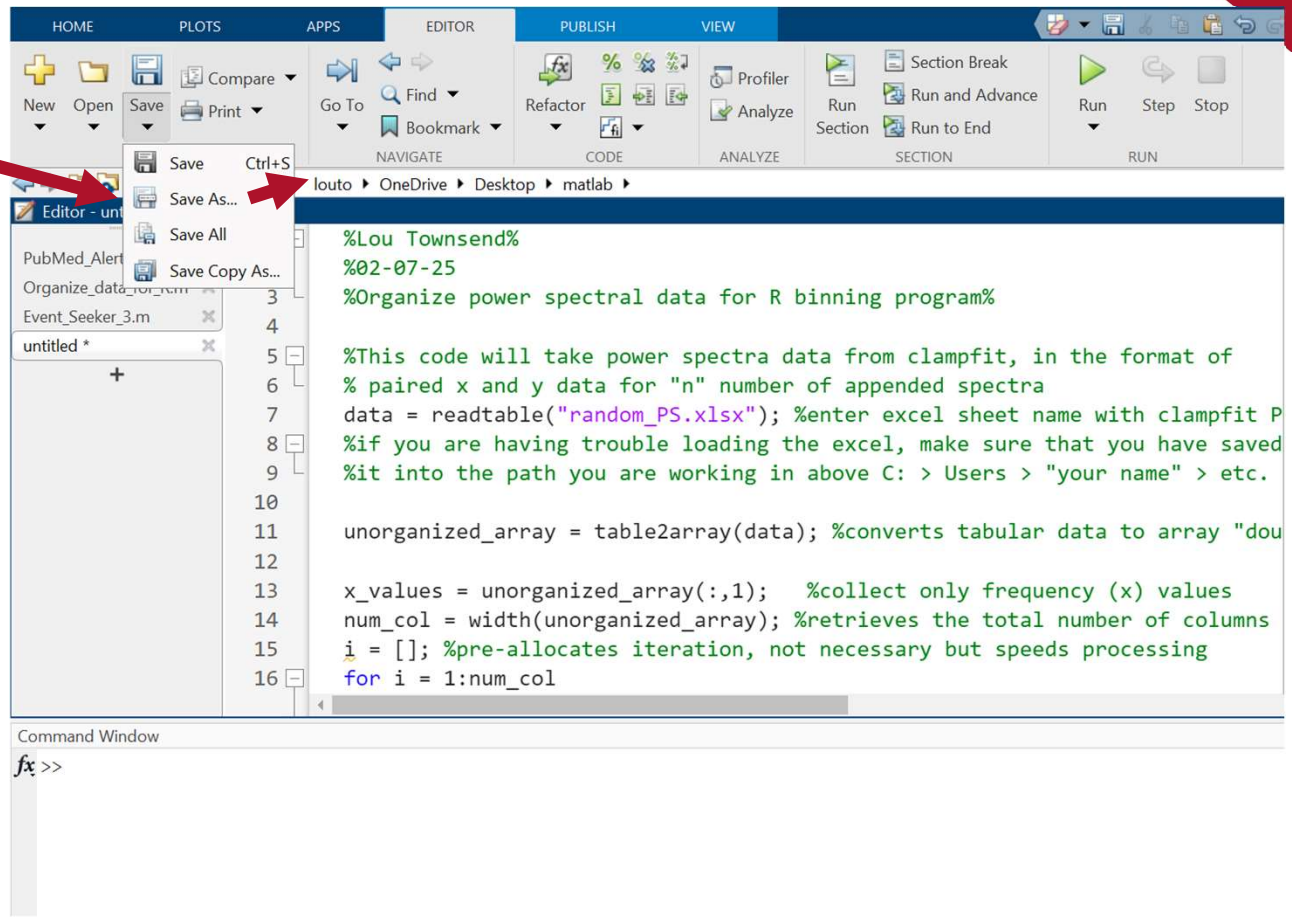
1. Work in the "Editor" tab
2. After clicking New to create a new script in the Editor, "untitled.m" will appear
3. Paste the code from the "Lou_MATLAB.txt" file into the blank space and save the script



Using MATLAB and loading excel files containing data

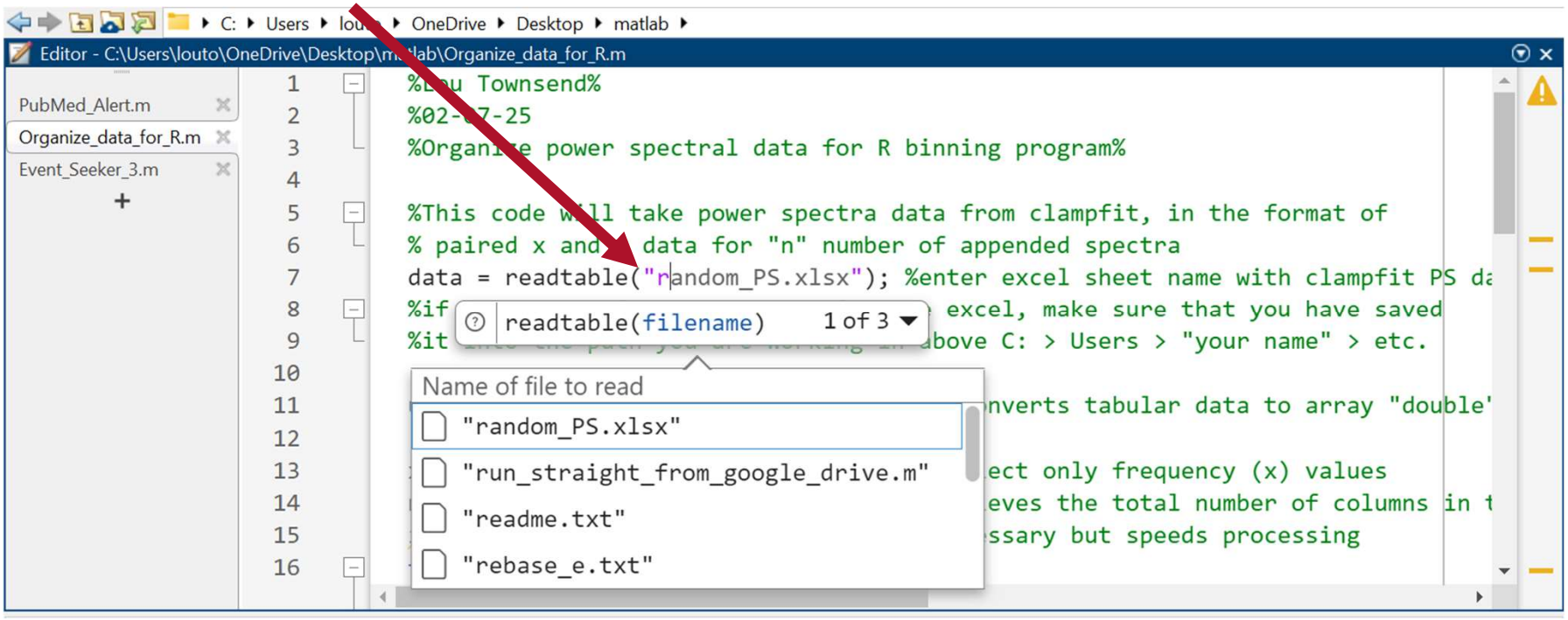
1. Save the script in the path with the folder you intend to work out of. Mine is “C: Users > louto > OneDrive > Desktop > matlab” just like before

2. I have called it “Organize_data_for_R.m”



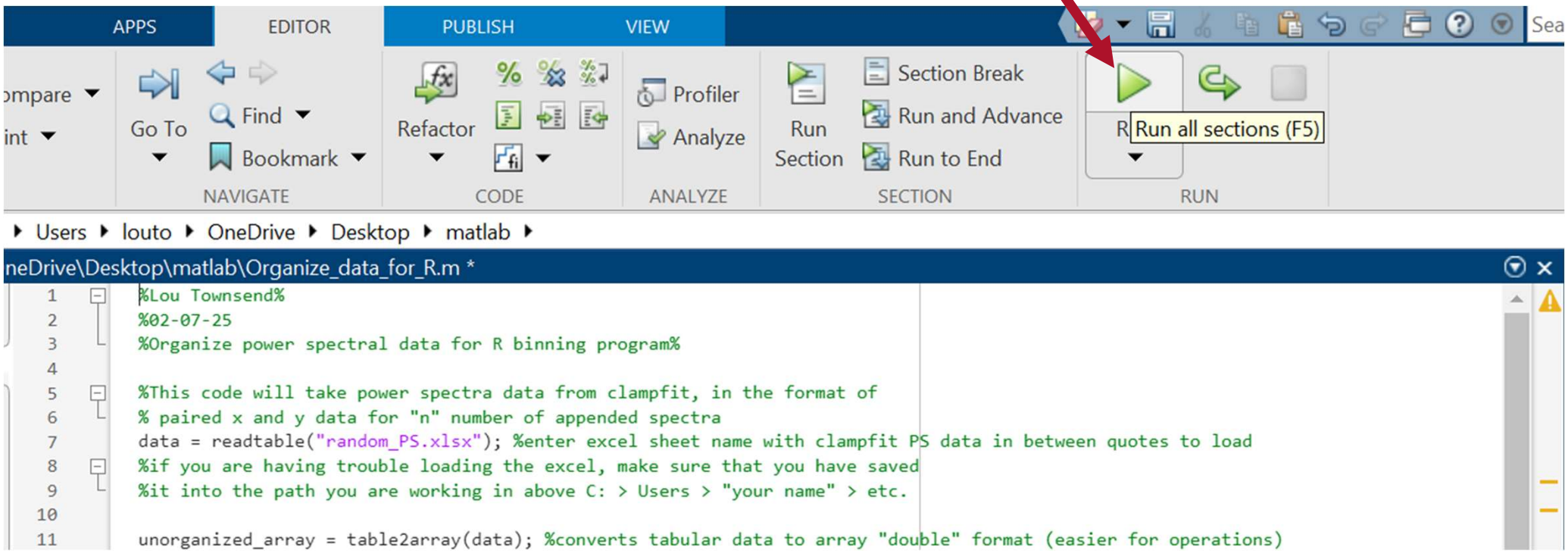
Using MATLAB and loading excel files containing data

1. Now that you have successfully saved your script, you can run your data through it
2. In line 7, you may search for your excel data by typing the first few letters of the file name, it will automatically search in your path folder (“matlab” in this example) to find this
3. Simply click on the Name of the file to read and it will autofill



Running the Program

1. Make sure your cursor | is at the top of the script in line 1, then click “Run”



Running the Program

1. As MATLAB runs, variables will propagate the “Workspace” and output will appear in the “Command Window”
2. Double-click the blue link under “Value” for the “table_data” variable

Command Window

```
         0     3.9151e+09
   0.54863        0.13697
    1.0973       0.070691
    1.6459       0.050784
    2.1945       0.033183

      :            :

    278.16       0.048906
     278.7        0.03903
    279.25       0.087518
     279.8       0.050099
    280.35       0.038197

  Display all 2560 rows.

fx >>
```

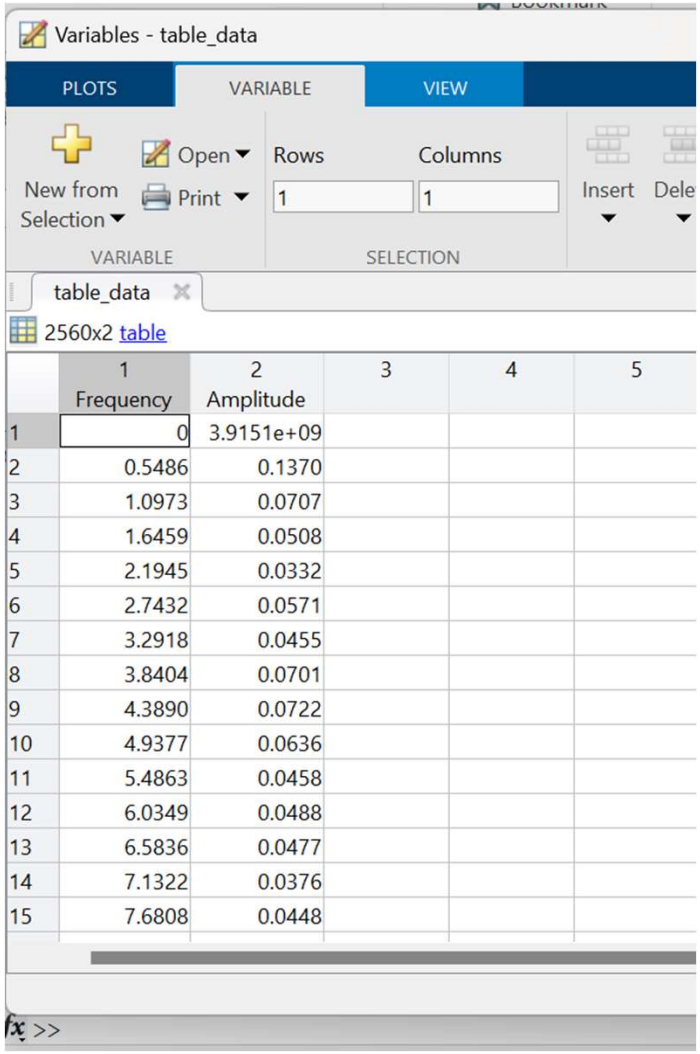
Workspace

Name	Value
Array_for...	512x6 double
col	5
data	512x10 table
format_bui...	2560x2 double
i	10
mm	6
num_col	10
only_ys	512x5 double
table_data	2560x2 table
unorganiz...	512x10 double
x_values	512x1 double

Running the Program

1. The “Variables” window will pop up showing your newly organized Frequency and Amplitude values

2. Woohoo!



	1 Frequency	2 Amplitude	3	4	5
1	0	3.9151e+09			
2	0.5486	0.1370			
3	1.0973	0.0707			
4	1.6459	0.0508			
5	2.1945	0.0332			
6	2.7432	0.0571			
7	3.2918	0.0455			
8	3.8404	0.0701			
9	4.3890	0.0722			
10	4.9377	0.0636			
11	5.4863	0.0458			
12	6.0349	0.0488			
13	6.5836	0.0477			
14	7.1322	0.0376			
15	7.6808	0.0448			

Exporting the new organized data

1. Highlight and copy the data from the variables window to a new excel sheet

2. I like to keep this excel sheet open as a working document, so I can keep adding in new power spectra data I am processing into the first two empty cells at the bottom of the sheet

3. You can combine several days of data together as well

Combining sheets from multiple scans

1. Make sure in your new excel file you have added column names “Frequency” and “Amplitude” if they did not paste well from Matlab

2. Now we can combine multiple days by highlighting each new organized result from Matlab output and pasting it in the two empty cells at the bottom of this working document

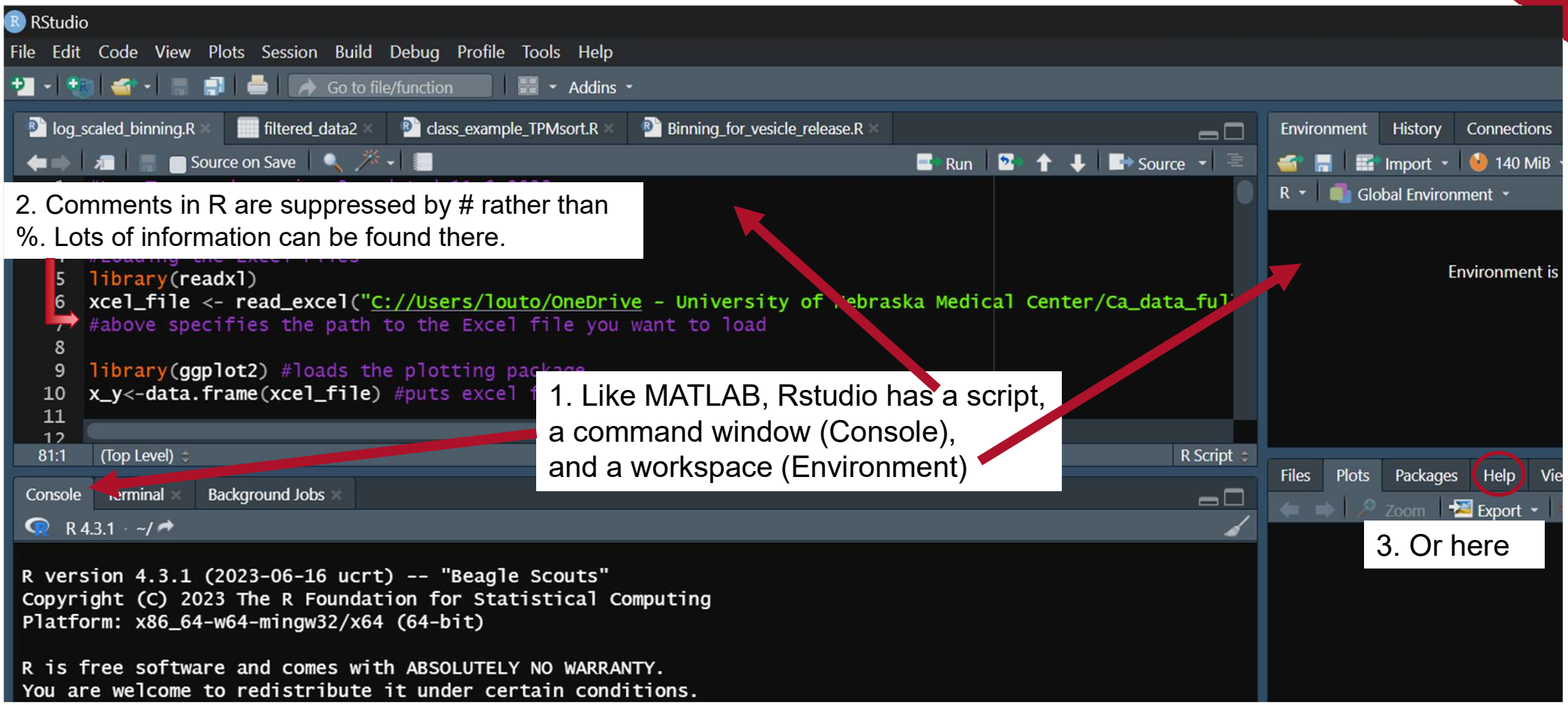
A1

	A	B
1	Frequency	Amplitude
2	0	3.92E+09
3	0.548631	0.136966
4	1.097261	0.070691
5	1.645892	0.050784
6	2.194523	0.033183
7	2.743153	0.057075
8	3.291784	0.045489
9	3.840415	0.070087
10	4.389045	0.072192
11	4.937676	0.063621

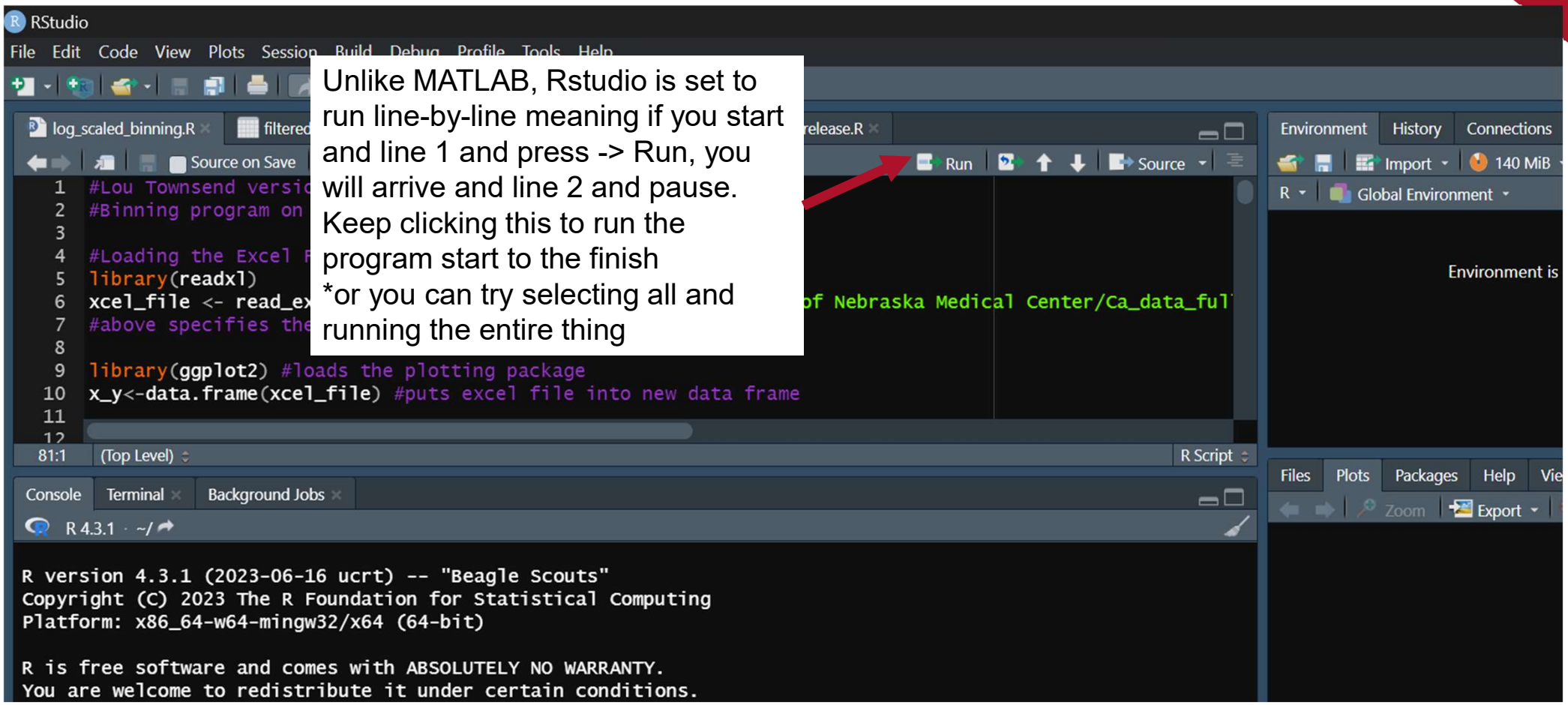
Paste next data set

	A	B
2554	276.5099	0.061962
2555	277.0585	0.103894
2556	277.6071	0.034196
2557	278.1557	0.048906
2558	278.7044	0.03903
2559	279.253	0.087518
2560	279.8016	0.050099
2561	280.3503	0.038197
2562		
2563		
2564		
2565		
2566		
2567		

Using R-RStudio



Using R-RStudio



Unlike MATLAB, Rstudio is set to run line-by-line meaning if you start and line 1 and press -> Run, you will arrive and line 2 and pause. Keep clicking this to run the program start to the finish

*or you can try selecting all and running the entire thing

Loading data into the binning program

Here I am loading a combined 61 cell calcium data sheet, “Ca_data_full…”, under my UNMC OneDrive and my user. These can be changed very easily depending on where you save your combined Excel files. Since I have tons in the “matlab” folder at this point, I can work out of this location by saving excel files there.

```
#Lou Townsend version 2 updat
#Binning program on R

#Loading the Excel Files
library(readxl)
xcel_file <- read_excel("C://Users/louto/OneDrive - University of Nebraska Medical Center/Ca_data_ful
#above specifies the path to the Excel file you want to load

library(ggplot2) #loads the plotting package
x_y<-data.frame(xcel_file)
```

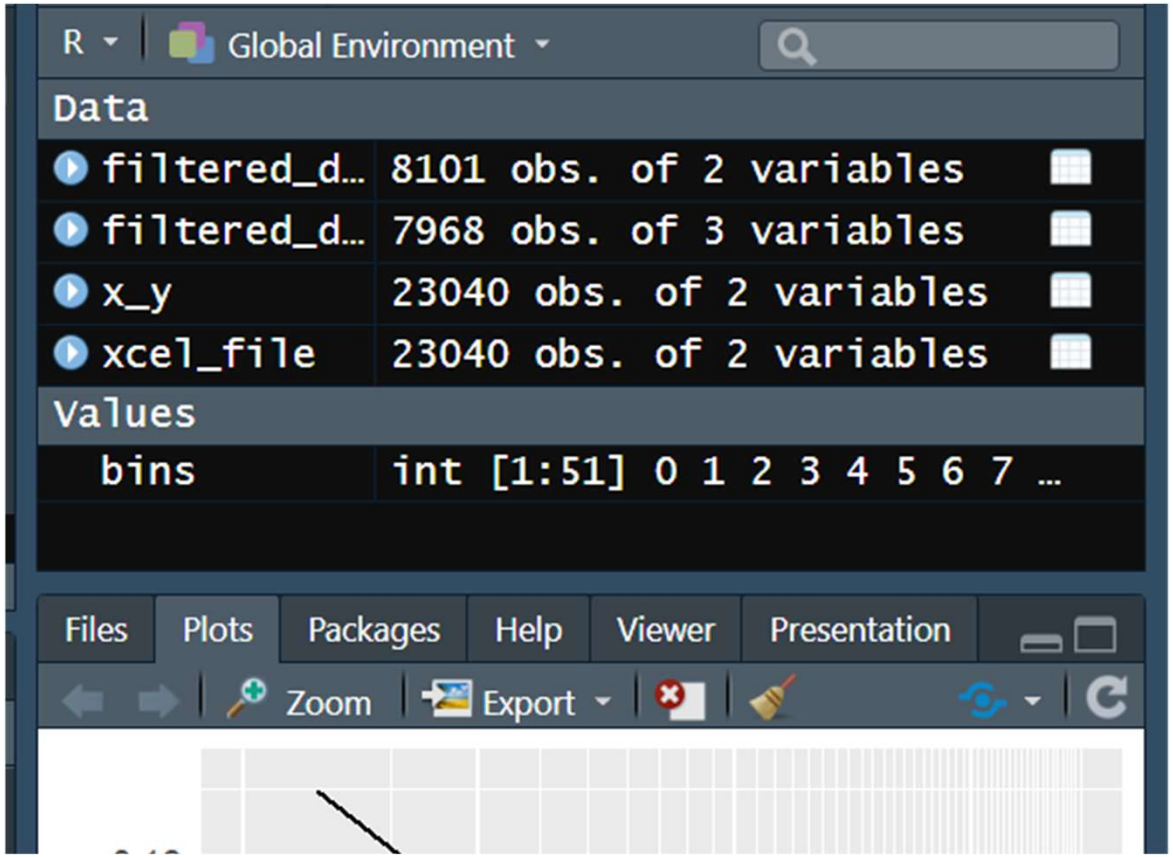
To load your combined, preformatted Excel file, you will search similarly in a path for your “folder”

```
R version 4.3.1 (2023-06-16 ucrt) -- "Beagle Scouts"
Copyright (C) 2023 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
```

Running the program

- As you run this program variables will appear one-by-one in the “Global Environment”
- A plot will appear under this environment
  - Instructions on scaling axis and limiting ranges embedded in the code

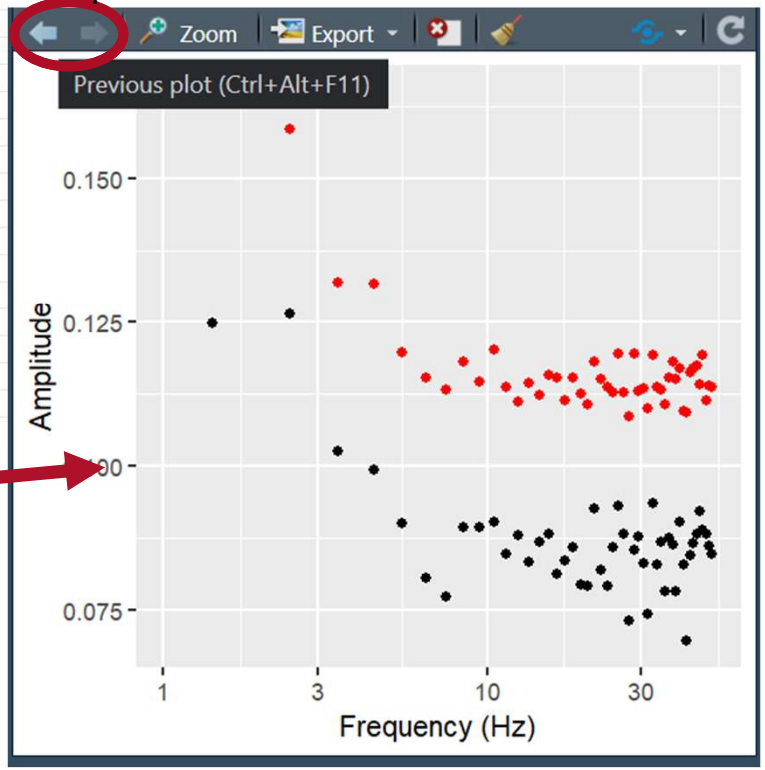


Exporting the data for further post-processing

- Once you reach the end, an excel window should emerge displaying your bins (centered at halfwidth), the std deviation per bin, and the mean power per bin as well
- This is helpful if you would like to process this data elsewhere rather than in R
- If you return to your R window, you should have two plots, one averaged power spectra, and one scatter plot with both std dev and mean (red)

	A	B	C	D	E
1		Frequency	Std Dev	Mean Power	
2	1	0.5	0.138569	0.192197	
3	2	1.5	0.124651	0.164839	
4	3	2.5	0.126277	0.158497	
5	4	3.5	0.102412	0.131938	
6	5	4.5	0.099265	0.131797	
7	6	5.5	0.090047	0.119615	
8	7	6.5	0.080552	0.115192	
9	8	7.5	0.077222	0.113242	
10	9	8.5	0.089298	0.118075	
11	10	9.5	0.089273	0.114416	
12	11	10.5	0.090142	0.119986	
13	12	11.5	0.084636	0.11352	
14	13	12.5	0.087878	0.111136	
15	14	13.5	0.083147	0.114324	
16	15	14.5	0.086796	0.112265	
17	16	15.5	0.088171	0.115714	
18	17	16.5	0.081082	0.115255	
19	18	17.5	0.083381	0.111377	
20	19	18.5	0.085714	0.115273	
21	20	19.5	0.079212	0.112547	

Toggle between plots



Good luck with your data!